

SIMATS ENGINEERING
(SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2

Course Code:	ITA0609	Course Title:	Machine Learning
CO Assessed:	CO 2 – Analyze (BL 3)	Assessment:	Assessment Tool 3
Weightage:	40% of CIA	Total Marks:	30 (3 Questions ×10 Mark)
CO1 Statement:	<i>To acquire a foundational understanding of key machine learning concepts including version spaces, candidate elimination, inductive bias, and heuristic search (BL1)</i>		
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Section / Batch:	SLOT A	Date:	27.7.26

Code of Conduct

I (N.kishan kumar-192424404) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Instructions.

- This test contains 5 Short questions, each carrying 10 mark.
- All questions are application-based and require analytical thinking and practical implementation. · No negative marking. Unanswered questions carry zero marks.
- Provide suitable algorithms, logic, calculations, or case-based solutions wherever required.

1. Backpropagation Algorithm in Multilayer Perceptron

Introduction

A Multilayer Perceptron (MLP) is a type of neural network containing an input layer, one or more hidden layers, and an output layer. For binary classification, the output is generally represented as **0 or 1**. Backpropagation is used to train the network by adjusting its weights to reduce prediction error.

Steps in Backpropagation

1. Forward Pass:

The input values are passed from the input layer to the hidden layer and then to the output layer. Each neuron calculates a weighted sum:

$$z = (w_1x_1 + w_2x_2 + \dots + b)$$

An activation function is applied to produce the neuron output.

2. Error Calculation:

The predicted output is compared with the actual output. For example:

Actual Output = 1

Predicted Output = 0.7

Error = $1 - 0.7 = 0.3$

The error shows how far the prediction is from the expected result.

3. Backward Propagation:

The error is propagated backward from the output layer toward the hidden layers. The gradients of the error with respect to each weight are calculated.

4. Weight Update:

The weights are updated using:

$$\text{New Weight} = \text{Old Weight} - \text{Learning Rate} \times \text{Gradient}$$

This process is repeated for many training iterations until the error becomes small.

Simple Example

Consider two input values:

$x_1 = 0.5, x_2 = 0.8$

Actual Output = 1

During the forward pass, suppose the neural network predicts **0.6**.

Error = $1 - 0.6 = 0.4$

Backpropagation calculates how much each weight contributed to this error. The weights are then adjusted, and during the next forward pass, the prediction may improve from **0.6 to 0.75**, moving closer to the actual output of 1.

Learning Rate and Activation Functions

The **learning rate** controls how much the weights change during each update. A very high learning rate may cause unstable training, while a very low learning rate makes training slow.

Activation functions such as **Sigmoid, ReLU, and Tanh** introduce non-linearity into the network. For binary classification, the **Sigmoid function** is commonly used in the output layer because it produces values between 0 and 1.

Conclusion

Backpropagation improves the neural network by repeatedly performing forward propagation, calculating errors, propagating those errors backward, and updating the weights. Proper learning rate and activation functions help the model achieve better convergence and classification accuracy.

2. Genetic Algorithm for Speed and Accuracy Optimization

Introduction

A Genetic Algorithm (GA) is an optimization technique inspired by natural evolution. It starts with a population of candidate solutions and improves them through selection, crossover, and mutation.

For this example, the objective is to identify a solution having both high speed and high accuracy.

Sample Dataset

Solution	Speed	Accuracy	Suitable
S1	60	70%	No
S2	75	80%	Yes
S3	50	65%	No
S4	85	90%	Yes
S5	70	75%	No

A simple fitness function can be defined as:

$$\text{Fitness} = 0.5 \times \text{Speed} + 0.5 \times \text{Accuracy}$$

Therefore:

$$S1 = 65$$

$$S2 = 77.5$$

$$S3 = 57.5$$

$$S4 = \mathbf{87.5}$$

$$S5 = 72.5$$

The candidate with a higher fitness value is considered better.

Genetic Algorithm Operations

1. Selection:

The candidates with higher fitness values are selected as parents. Here, **S4 and S2** are selected because they have the highest fitness values.

$$S4 = (85 \text{ Speed}, 90\% \text{ Accuracy})$$

$$S2 = (75 \text{ Speed}, 80\% \text{ Accuracy})$$

2. Crossover:

Crossover combines characteristics of selected parents to generate new offspring.

For example:

$$\text{Parent 1} = (85, 90)$$

$$\text{Parent 2} = (75, 80)$$

A new offspring may be:

$$\mathbf{\text{Offspring} = (85, 80)}$$

or another combination such as:

$$\mathbf{\text{Offspring} = (75, 90)}$$

Thus, crossover generates new candidate solutions by combining good characteristics from existing solutions.

3. Mutation:

Mutation randomly changes a small part of an offspring. It helps the algorithm explore new solutions instead of repeatedly producing similar candidates.

For example:

$$\text{Before Mutation} = (85, 90)$$

$$\text{After Mutation} = (88, 92)$$

$$\text{Fitness} = 0.5(88) + 0.5(92) = 90$$

Therefore, the mutated solution has better fitness than the original candidates.

Evolution Across Generations

Generation	Best Speed	Best Accuracy	Fitness
Generation 1	85	90%	87.5
Generation 2	88	92%	90
Generation 3	90	94%	92

The quality of the population improves across generations.

Final Optimized Solution

After selection, crossover, and mutation for multiple generations:

Speed = 90

Accuracy = 94%

Fitness = 92

Suitable = Yes

Therefore, **(90 Speed, 94% Accuracy)** is the best candidate solution in this illustrative evolution.

Conclusion

Selection chooses candidates with better fitness, crossover combines useful characteristics from parents, and mutation introduces new variations. Together, these operations allow the Genetic Algorithm to search the solution space and gradually obtain an optimized solution.

3. Neural Network Classification and Performance Evaluation

Introduction

A neural network can be used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The model is trained using backpropagation to minimize the difference between actual and predicted outputs.

Dataset

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual	Predicted
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Training Using Backpropagation

First, patient attributes are given as inputs to the neural network. Categorical attributes such as Blood Pressure and Lifestyle Risk are converted into numerical values. The network performs a forward pass and predicts whether the patient has the disease. The predicted result is compared with the actual result to calculate the error. Backpropagation sends this error backward through the network and adjusts the weights. This process is repeated until the network produces more accurate predictions.

Confusion Matrix Calculation

From the given dataset:

Example 1: Actual Yes, Predicted Yes → True Positive (TP)

Example 2: Actual Yes, Predicted No → False Negative (FN)

Example 3: Actual No, Predicted No → True Negative (TN)

Example 4: Actual Yes, Predicted Yes → True Positive (TP)

Example 5: Actual No, Predicted Yes → False Positive (FP)

Therefore:

True Positive (TP) = 2

True Negative (TN) = 1

False Positive (FP) = 1

False Negative (FN) = 1

Confusion Matrix

	Predicted Yes	Predicted No
Actual Yes	TP = 2	FN = 1
Actual No	FP = 1	TN = 1

Evaluation Metrics

1. Accuracy

Accuracy = (TP + TN) / Total

= (2 + 1) / 5

= 3/5 = 60%

Therefore, the model correctly predicts 60% of the examples.

2. Precision

Precision = TP / (TP + FP)

= 2 / (2 + 1)

= **66.67%**

This means that among the patients predicted as having the disease, approximately 66.67% were actually positive.

3. Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$= 2 / (2 + 1)$$

$$= \mathbf{66.67\%}$$

This means the model correctly identifies approximately 66.67% of actual disease-positive cases.

4. F1-Score

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

$$= \mathbf{66.67\%}$$

The F1-score provides a balance between precision and recall.

Preprocessing and Feature Selection

Preprocessing improves data quality before training. Missing values can be handled, numerical features such as Age can be normalized, and categorical values such as High, Normal, Medium, and Low can be converted into numerical form.

Feature selection identifies the most useful attributes for prediction. Removing irrelevant or redundant features can reduce model complexity, decrease training time, and improve generalization.

For example, if Glucose Level and Blood Pressure strongly influence disease prediction, these features may receive greater importance during model training.

Conclusion

The neural network uses backpropagation to learn patterns between patient characteristics and disease outcomes. For the given dataset, TP = 2, TN = 1, FP = 1, and FN = 1, resulting in 60% accuracy, 66.67% precision, 66.67% recall, and 66.67% F1-score. Proper preprocessing, feature selection, and training can further improve the model's accuracy and reliability.